# ADESH KUMAR SAHOO

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## SUMMARY

Applied Machine Learning and Natural Language Processing practitioner with hands-on experience in building retrieval-based NLP systems, predictive ML pipelines, and LLM evaluation frameworks. Experienced in modular ML workflows, model evaluation, and real-world deployment.

## PROJECTS

**LLM and RAG Evaluation Benchmarking Framework**
*Sentence Transformers, OpenAI API, FastAPI, Streamlit*

- Designed and deployed a benchmarking framework for comparing baseline LLM and RAG-based systems for QA evaluation using semantic retrieval and LLM-based generation.
- Built modular retrieval, generation, evaluation, and orchestration pipelines supporting both local and cloud inference workflows.
- Implemented multi-metric benchmarking using Exact Match, Semantic Similarity, Recall@k, MRR, and LLM-as-Judge evaluation.
- Improved semantic similarity by 58% and LLM-judged correctness by 35.8% over baseline responses across a 150-sample benchmark, with 95% Recall@k retrieval performance.

**End-to-End Flight Delay Prediction System - Predicting Risk, Severity, and Causes using Pre-departure Information**
*Scikit-learn, XGBoost, Streamlit, FastAPI*

- Designed and deployed an end-to-end flight delay prediction system enabling real-time pre-departure prediction of delay risk and severity through a modular ML pipeline.
- Engineered production-oriented ML inference workflows by resolving training-serving skew, fixing feature-space inconsistencies, and deployment memory constraints, ensuring stable predictions across local and cloud environments.
- Built an aviation-informed multitask prediction system for delay risk, severity, and possible causes using domain-specific delay logic while enforcing strict data-leakage prevention and reusable preprocessing pipelines.
- Evaluated multiple modeling approaches and identified feature richness over model complexity as the primary performance bottleneck, informing future feature-engineering and data expansion efforts.

**Context-Aware Business Japanese to English Interpreter Assistant (Ongoing Project)**

- Currently building a manually curated bilingual business Japanese–English corpus (~500 sentence pairs) sourced from real business communication and annotated with metadata such as scenario, domain, and formality.
- Developing a context-aware business communication assistant using multilingual retrieval, metadata-aware ranking, and human-in-the-loop suggestion selection.

## SKILLS

Languages: Python
Machine Learning: Classification, Regression, Feature Engineering, Model Evaluation, Pipelines
NLP and LLMs: Sentence Transformers, Semantic Retrieval, RAG, Embeddings
Libraries: Scikit-learn, Numpy, Pandas
Deployment: Streamlit, FastAPI, Render
Tools: Git, Jupyter

## EDUCATION

Bachelor of Technology in Information Technology
Meghnad Saha Institute of Technology
2020 - 2024 | CGPA: 8.79

## CERTIFICATIONS

Deep Learning Specialization - Andrew Ng (Coursera)
Machine Learning Specialization - Andrew Ng (Coursera)
Japanese Language Proficiency Test N2

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